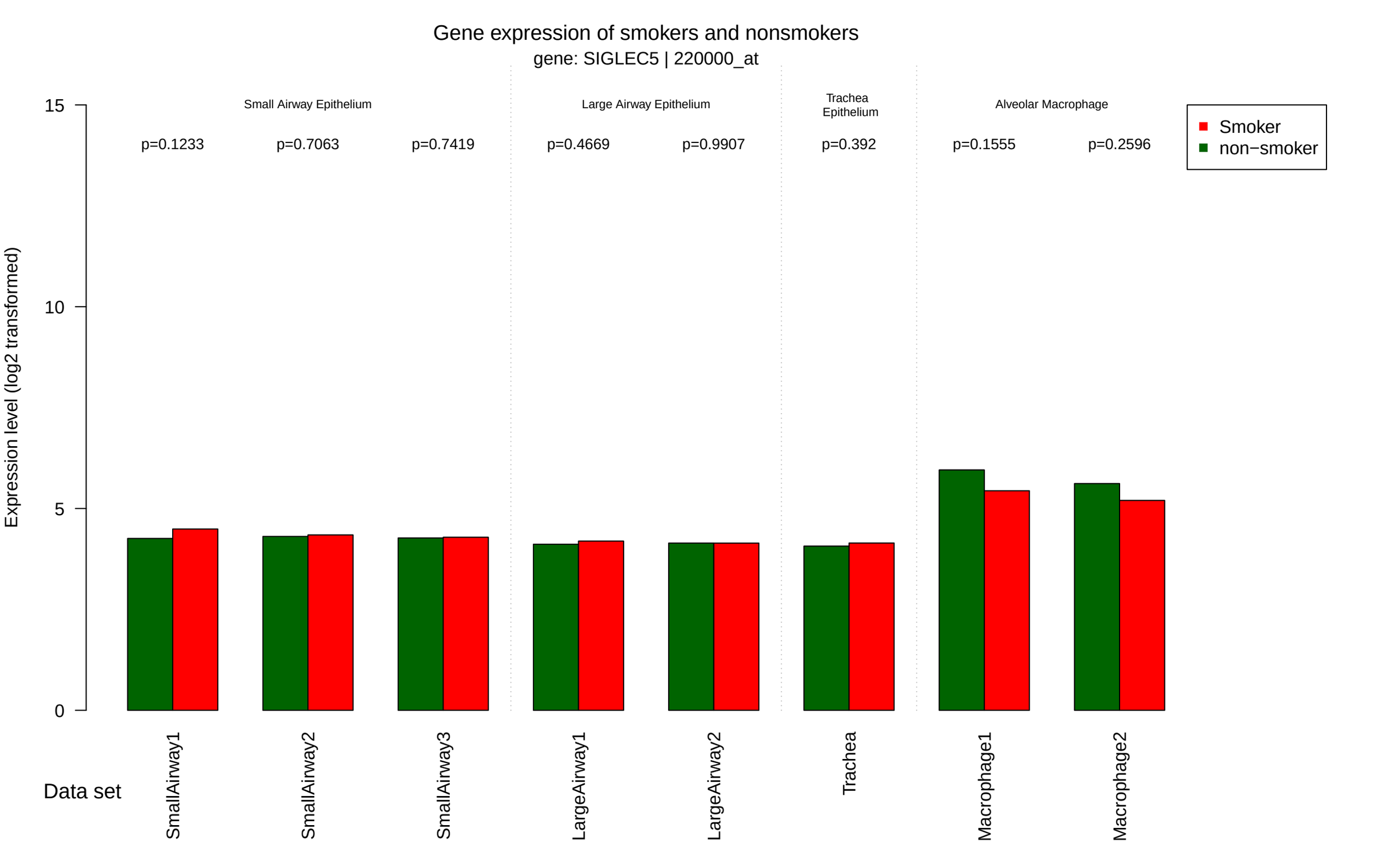
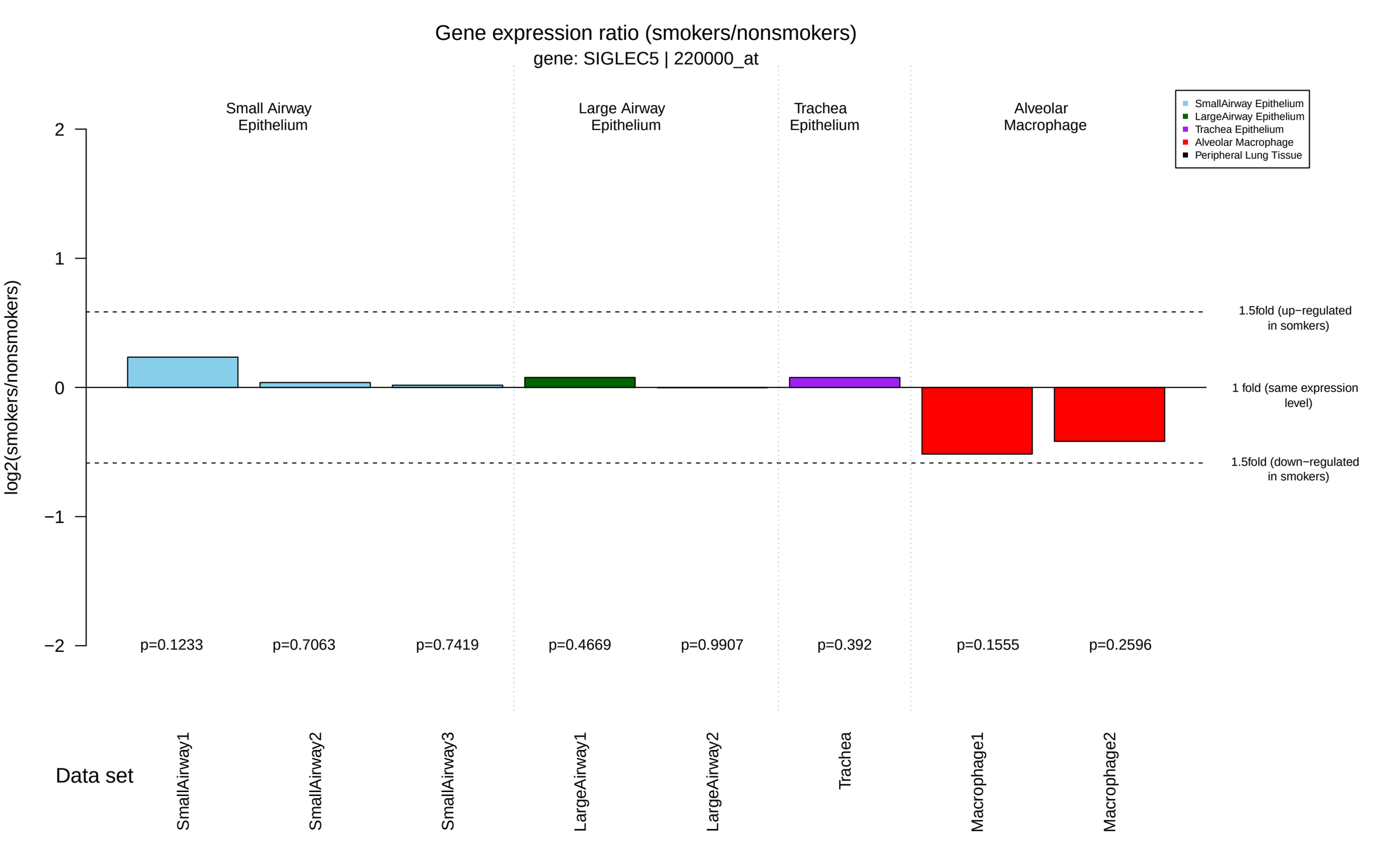
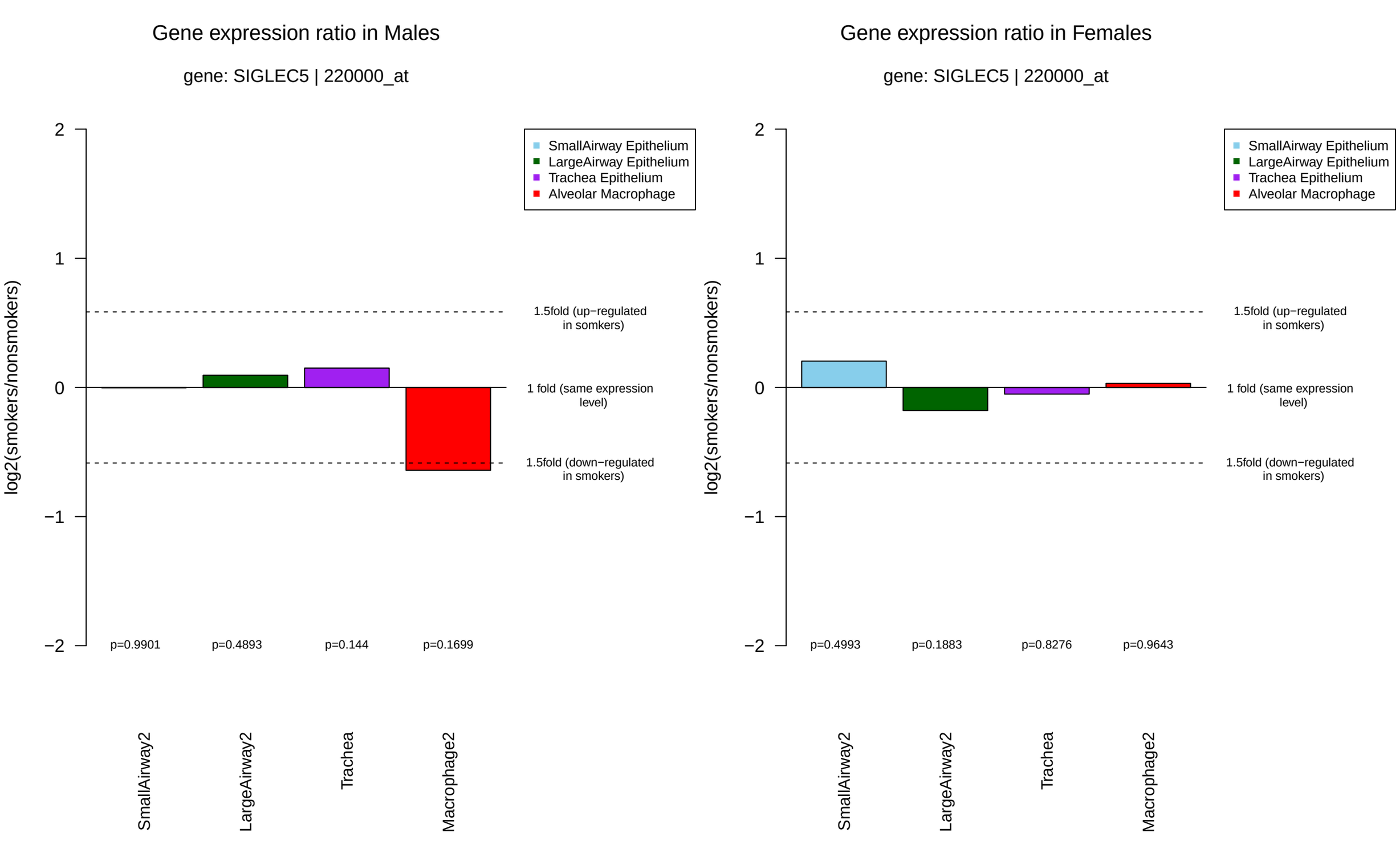
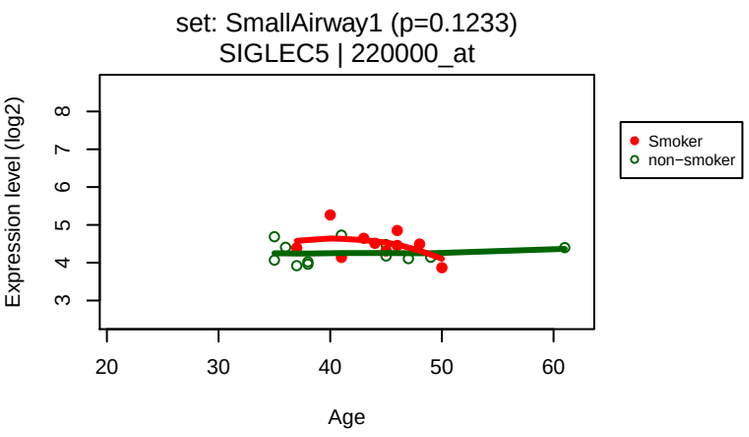
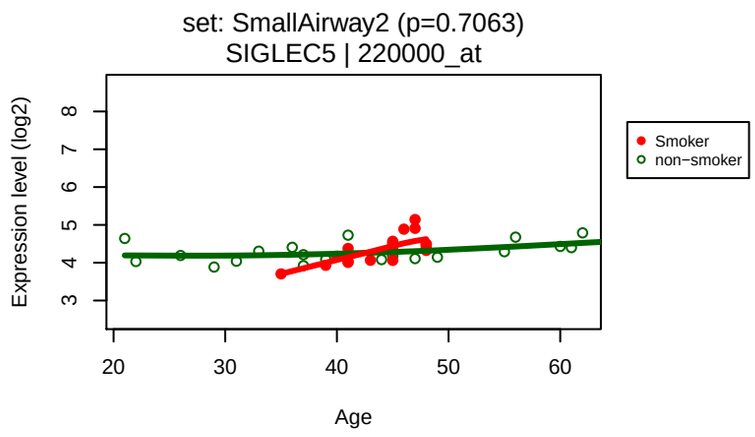
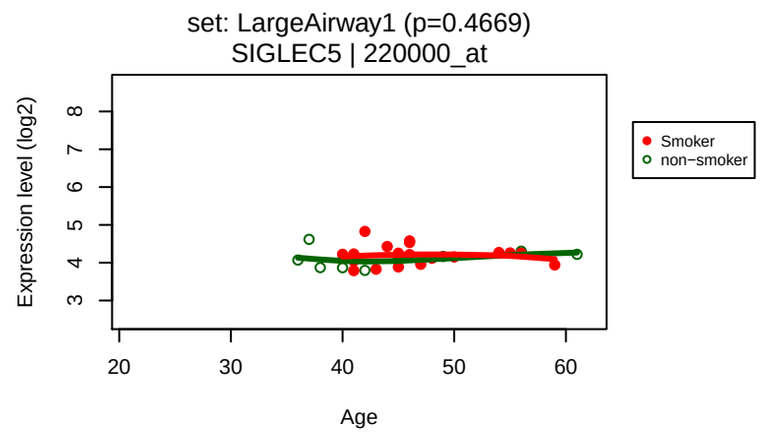

Gene expression of smokers and nonsmokers
gene: SIGLEC5 | 220000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1233
p=0.7063
p=0.7419
p=0.4669
p=0.9907
p=0.392
p=0.1555
p=0.2596
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

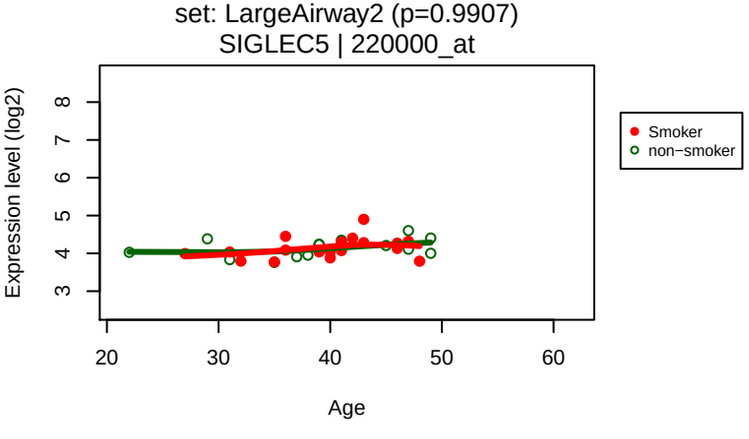

Gene expression ratio (smokers/nonsmokers)
gene: SIGLEC5 | 220000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1233
p=0.7063
p=0.7419
p=0.4669
p=0.9907
p=0.392
p=0.1555
p=0.2596
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

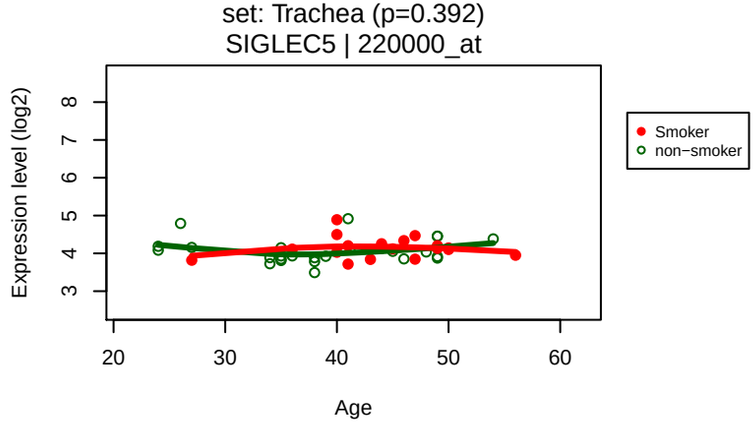

Gene expression ratio in Males
gene: SIGLEC5 | 220000_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9901
p=0.4893
p=0.144
p=0.1699
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIGLEC5 | 220000_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4993
p=0.1883
p=0.8276
p=0.9643
SmallAirway2
LargeAirway2
Trachea
Macrophage2

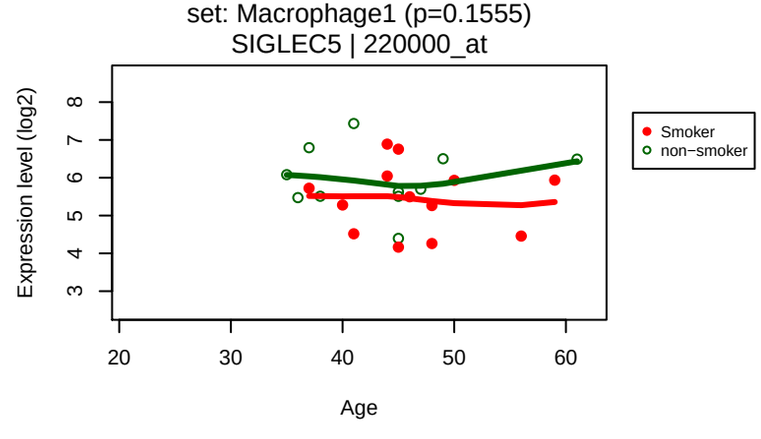

set: SmallAirway1 (p=0.1233)
SIGLEC5 | 220000_at
Expression level (log2)
Age
Smoker
non−smoker

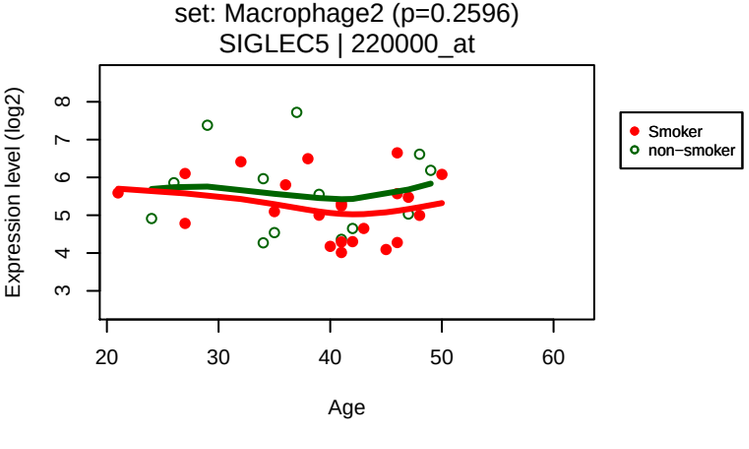

set: SmallAirway2 (p=0.7063)
SIGLEC5 | 220000_at
Expression level (log2)
Age
Smoker
non−smoker

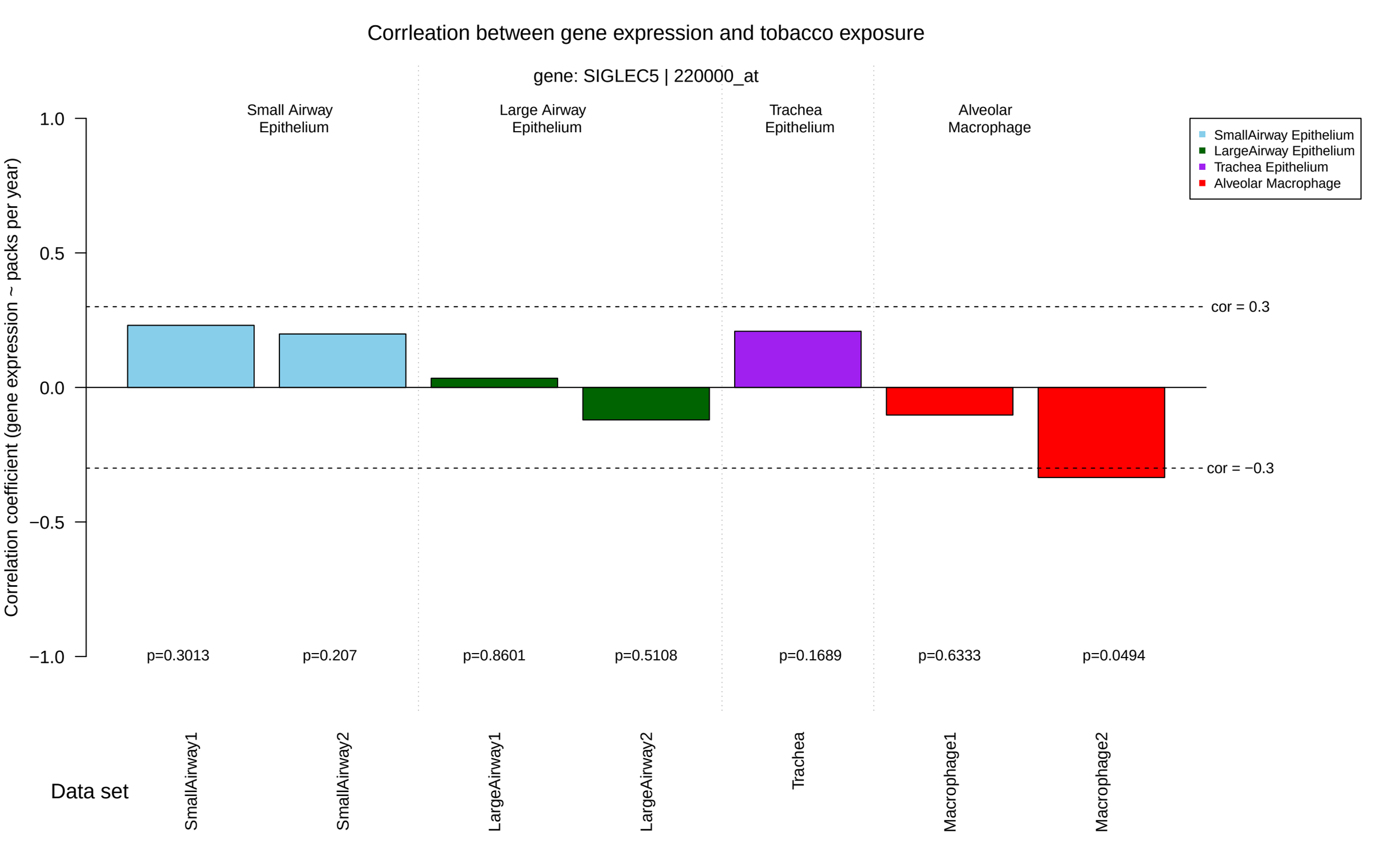

set: LargeAirway1 (p=0.4669)
SIGLEC5 | 220000_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9907)
SIGLEC5 | 220000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.392)
SIGLEC5 | 220000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1555)
SIGLEC5 | 220000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2596)
SIGLEC5 | 220000_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIGLEC5 | 220000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3013
p=0.207
p=0.8601
p=0.5108
p=0.1689
p=0.6333
p=0.0494
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2